CAD package and SOLIDWORKS workflow

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

Open SOLIDWORKS > File > Open > select STEP > cad/FilterSense_assembly.step. Open parts individually to inspect imported solids. Save As native documents if needed. Import was not run here; installed version is unknown.

For native feature trees: File > New > Part > set MMGS (bottom-right units). Select the named plane > Sketch. Use Smart Dimension to fully define the profile, then Features > Extruded Boss/Base or Extruded Cut as listed on each sheet.

Assembly: File > New > Assembly > insert stand first and fix it. Insert each part. Use Concentric mates for housing/cartridge axes, Coincident mates for mounting faces, and Distance mates for height/offset. The manifest gives assumed assembly origins in mm. STEP positions do not imply mates.

Exploded view: Assembly tab > Exploded View; select cartridges/caps/lid and translate upward about 140 mm; move housings forward 65 mm. An exported exploded STEP is supplied. Clearance and supports must be refined against real components.

Engineering drawing: File > Make Drawing from Part/Assembly > choose sheet > Model View > front/top/right; use Smart Dimension or Model Items for native features. These reference sheets give actual projections, overall envelopes and specified sketch dimensions, not production tolerances.

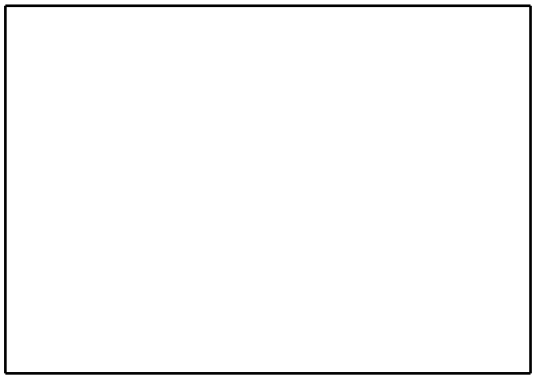
Feature notes: Boss adds material; Cut removes it. Revolve rotates a closed half-section about a centreline (alternative for axisymmetric housings). Hole Wizard defines standard holes. Fillet rounds edges; Chamfer bevels them. No decorative fillets are applied here; mating/pressure details must be reviewed first.

No flow or structural simulation has been run. Check Tools > Add-Ins and licence availability before using simulation. A Premium licence alone does not establish Flow Simulation entitlement.

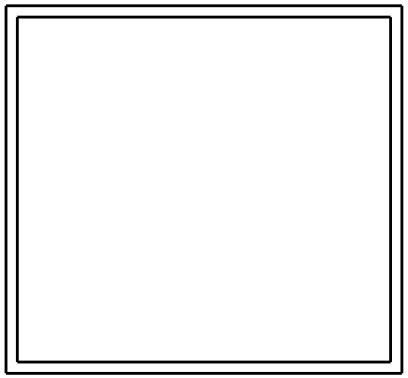
01 feed tank

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 140 mm Y = 130 mm Z = 200 mm

Purpose: Feed storage. Assembly origin: [-235, 10, 12] mm.

Native feature sequence: Top Plane: centre rectangle 140 x 130; Boss-Extrude 200. Select top face, Sketch centre rectangle 132 x 122; Cut-Extrude 196. Wall/base 4.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

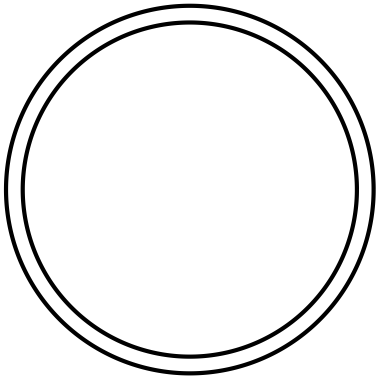
02 sediment housing

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 88 mm Y = 88 mm Z = 270 mm

Purpose: Sediment housing envelope. Assembly origin: [-65, 0, 40] mm.

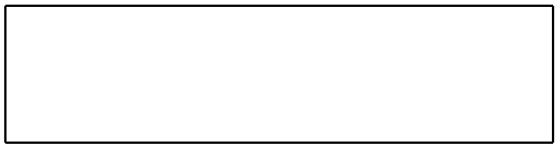
Native feature sequence: Top Plane: circle diameter 88; Boss-Extrude 270. Top face: circle diameter 80; Cut-Extrude 266. Wall/base 4. No threaded closure or seal is designed.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

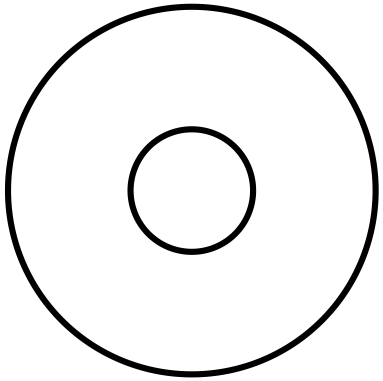
03 sediment cartridge

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 60 mm Y = 60 mm Z = 240 mm

Purpose: Sediment cartridge envelope. Assembly origin: [-65, 0, 48] mm.

Native feature sequence: Top Plane: concentric circles diameter 60 and 20; Boss-Extrude 240. Assumed annular cartridge; filtration media not modelled.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

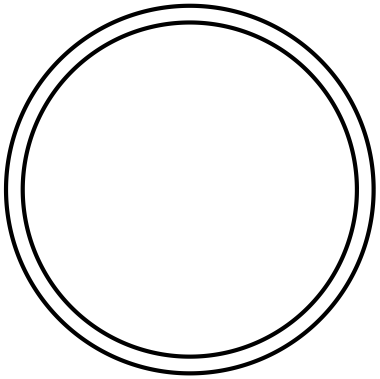
04 carbon housing

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 88 mm Y = 88 mm Z = 270 mm

Purpose: Carbon housing envelope. Assembly origin: [55, 0, 40] mm.

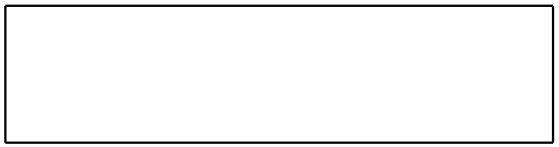
Native feature sequence: Repeat sediment housing: diameter 88 x 270; bore diameter 80, depth 266. Save as a separate part.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

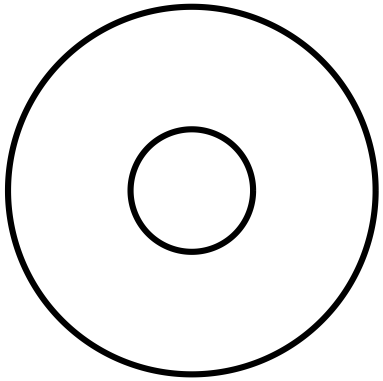
05 carbon cartridge

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 60 mm Y = 60 mm Z = 240 mm

Purpose: Carbon cartridge envelope. Assembly origin: [55, 0, 48] mm.

Native feature sequence: Top Plane: circles diameter 60 and 20; Boss-Extrude 240. Carbon media and adsorption life are not modelled.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

06 ro housing

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 310 mm Y = 56 mm Z = 56 mm

Purpose: RO housing envelope. Assembly origin: [-145, 25, 390] mm.

Native feature sequence: Right Plane: circle diameter 56; Boss-Extrude 310. End face: circle diameter 50; Cut-Extrude 307. Axis along X. No pressure-rated closure.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

07 ro membrane

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 280 mm Y = 44 mm Z = 44 mm

Purpose: RO membrane envelope. Assembly origin: [-135, 25, 390] mm.

Native feature sequence: Right Plane: circles diameter 44 and 14; Boss-Extrude 280. Membrane layers and hydraulic internals omitted.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

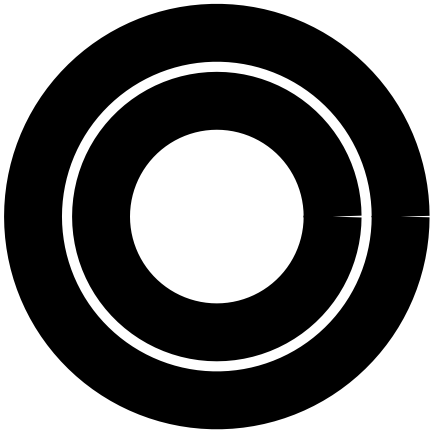
08 tube segment

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 6.35 mm Y = 6.35 mm Z = 100 mm

Purpose: Tube routing section. Assembly origin: [210, 0, 30] mm.

Native feature sequence: Top Plane: circles diameter 6.35 and 4; Boss-Extrude 100. For curved routing create a path sketch and Swept Boss/Base using this annular profile.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

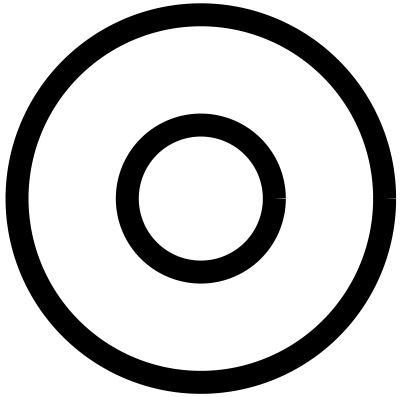
09 connector

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 16 mm Y = 16 mm Z = 24 mm

Purpose: Connector envelope. Assembly origin: [210, 0, 130] mm.

Native feature sequence: Top Plane: circles diameter 16 and 6.4; Boss-Extrude 24. Real connector thread, collet, seals and bore must follow vendor dimensions.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

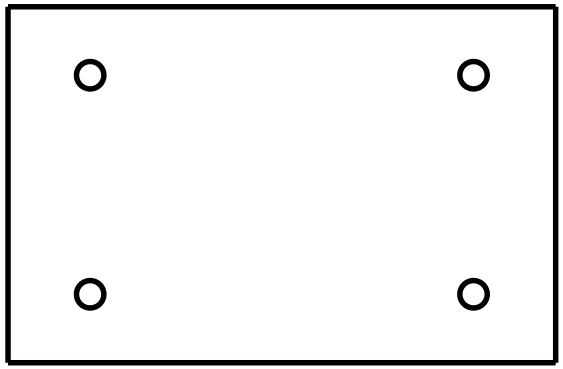
10 pump bracket

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 100 mm Y = 65 mm Z = 5 mm

Purpose: Pump mounting plate. Assembly origin: [215, 70, 12] mm.

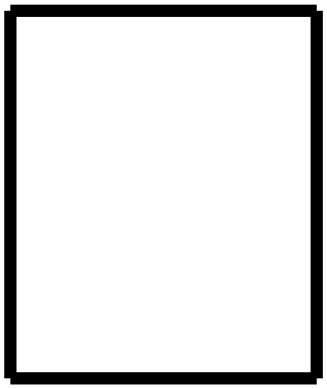
Native feature sequence: Top Plane: centre rectangle 100 x 65; Boss-Extrude 5. Top face: four diameter 5 circles centred at (+/-35, +/-20); Cut-Extrude Through All. Hole Wizard can replace cuts using plain holes.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

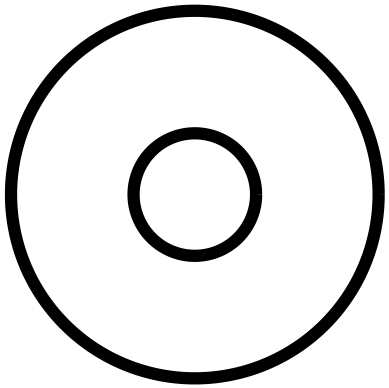
11 pressure sensor mount

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 30 mm Y = 30 mm Z = 25 mm

Purpose: Pressure sensor adapter envelope. Assembly origin: [-125, 0, 300] mm.

Native feature sequence: Top Plane: circles diameter 30 and 10; Boss-Extrude 25. Repeat as P1/P2 locations after matching actual transducer thread and pressure rating.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

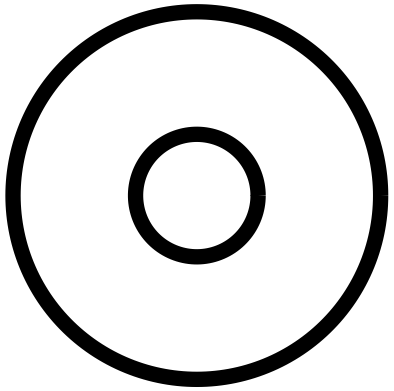
12 tds sensor mount

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 24 mm Y = 24 mm Z = 30 mm

Purpose: TDS probe holder envelope. Assembly origin: [-220, 15, 215] mm.

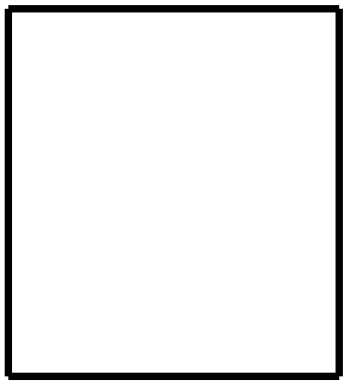
Native feature sequence: Top Plane: circles diameter 24 and 8; Boss-Extrude 30. Repeat for product probe after vendor dimensions and wet materials are checked.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

13 turbidity mount

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 50 mm Y = 40 mm Z = 45 mm

Purpose: Turbidity module bracket. Assembly origin: [-220, 60, 215] mm.

Native feature sequence: Top Plane: rectangle 50 x 40; Boss-Extrude 45. Top face: rectangle 44 x 34; Cut-Extrude 42. This is a mounting envelope, not a calibrated optical cell.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

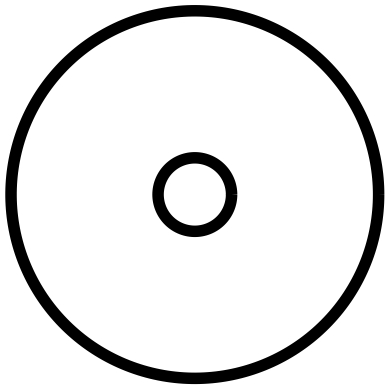
14 flow sensor mount

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 32 mm Y = 32 mm Z = 45 mm

Purpose: Flow sensor envelope. Assembly origin: [210, 0, 160] mm.

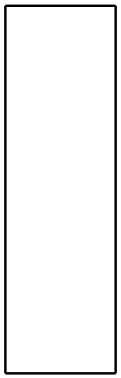
Native feature sequence: Top Plane: circles diameter 32 and 6.4; Boss-Extrude 45. Adapt to a meter rated for the low product flow; calibrate pulses/litre.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

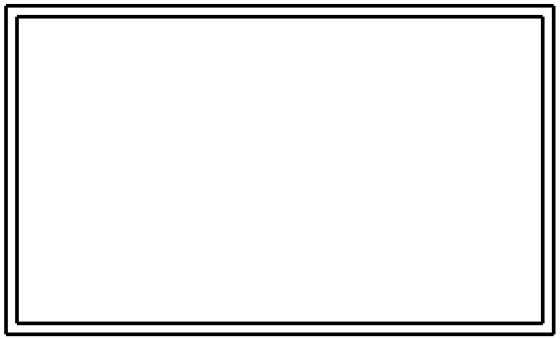
15 electronics enclosure

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 150 mm Y = 90 mm Z = 45 mm

Purpose: Dry electronics enclosure. Assembly origin: [210, 65, 440] mm.

Native feature sequence: Top Plane: rectangle 150 x 90; Boss-Extrude 45. Top face: rectangle 144 x 84; Cut-Extrude 42. Keep cable entries above drip zone; ingress protection is unverified.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

16 electronics plate

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT

TOP



Overall X = 140 mm Y = 80 mm Z = 3 mm

Purpose: Electronics mounting plate. Assembly origin: [210, 65, 444] mm.

Native feature sequence: Top Plane: rectangle 140 x 80; Boss-Extrude 3. Four diameter 3 through holes centred (+/-60,+/-30). PCB spacers remain vendor dependent.

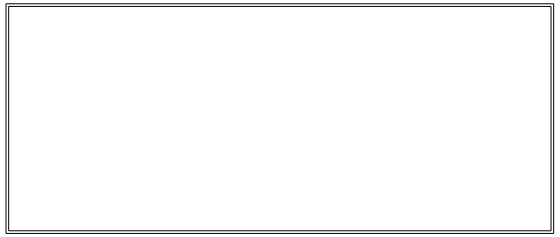
Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

17 leak tray

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT

TOP



Overall X = 620 mm Y = 260 mm Z = 12 mm

Purpose: Leak collection tray. Assembly origin: [0, 0, 0] mm.

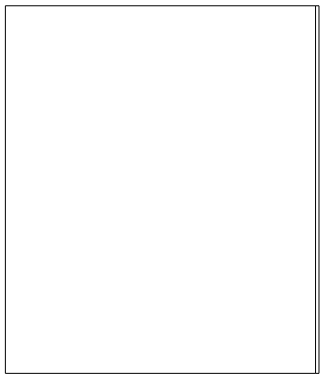
Native feature sequence: Top Plane: rectangle 620 x 260; Boss-Extrude 12. Top face: rectangle 614 x 254; Cut-Extrude 9. Add approved leak sensor at lowest point in detailed design.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

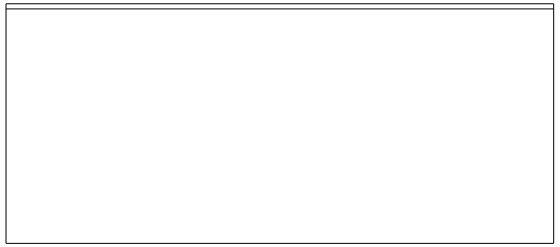
18 stand

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 640 mm Y = 280 mm Z = 546 mm

Purpose: Prototype stand envelope. Assembly origin: [0, 0, 0] mm.

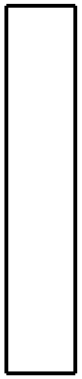
Native feature sequence: Top Plane: rectangle 640 x 280; Boss-Extrude 6 downward. On top at rear edge: rectangle 640 x 6 centred at Y=137; Boss-Extrude 540. Merge solids. This is a conceptual backplate; stiffness/fasteners not verified.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

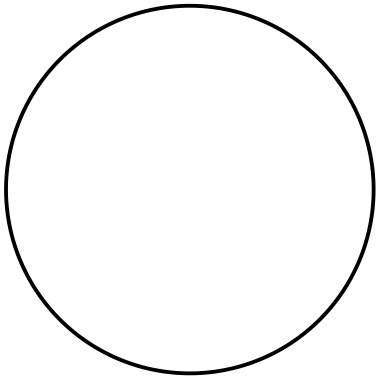
19 sediment cap

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 96 mm Y = 96 mm Z = 18 mm

Purpose: Sediment cap envelope. Assembly origin: [-65, 0, 310] mm.

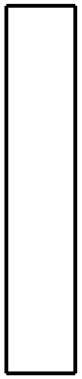
Native feature sequence: Top Plane: circle diameter 96; Boss-Extrude 18. No threaded or sealed interface is provided.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

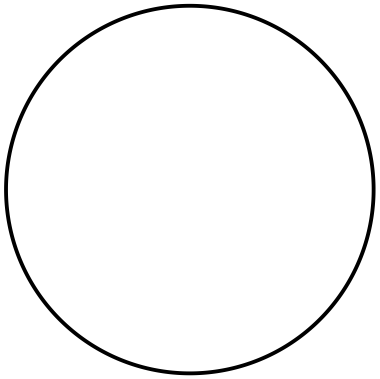
20 carbon cap

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 96 mm Y = 96 mm Z = 18 mm

Purpose: Carbon cap envelope. Assembly origin: [55, 0, 310] mm.

Native feature sequence: Top Plane: circle diameter 96; Boss-Extrude 18. No threaded or sealed interface is provided.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

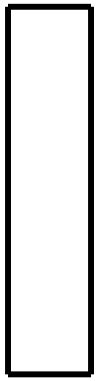
21 ro cap

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 14 mm Y = 62 mm Z = 62 mm

Purpose: RO cap envelope. Assembly origin: [165, 25, 390] mm.

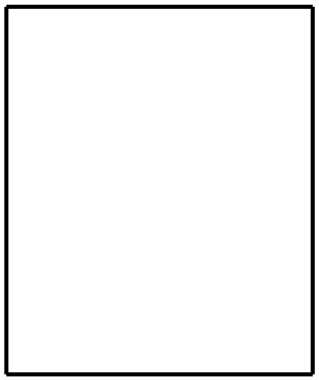
Native feature sequence: Right Plane: circle diameter 62; Boss-Extrude 14. No threaded or sealed interface is provided.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

22 pump envelope

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 90 mm Y = 55 mm Z = 75 mm

Purpose: Pump clearance envelope. Assembly origin: [215, 70, 17] mm.

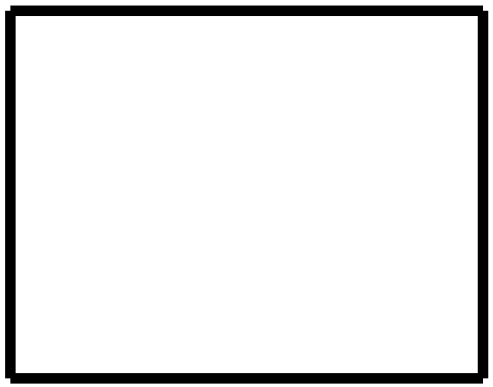
Native feature sequence: Top Plane: rectangle 90 x 55; Boss-Extrude 75. Replace with vendor geometry; this is not a pump mechanism.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

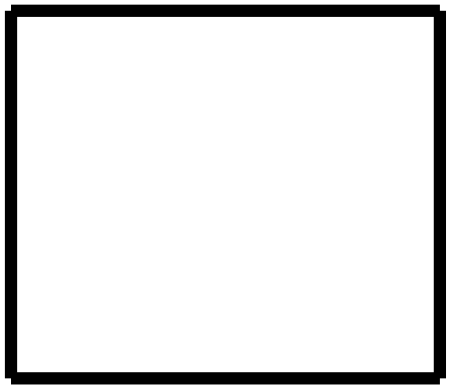
23 nc valve envelope

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT



TOP



Overall X = 35 mm Y = 30 mm Z = 45 mm

Purpose: Normally closed valve clearance. Assembly origin: [-150, -50, 20] mm.

Native feature sequence: Top Plane: rectangle 35 x 30; Boss-Extrude 45. Replace with rated valve geometry. Simulated shutdown does not prove valve behaviour.

Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

24 enclosure lid

CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

FRONT

TOP



Overall X = 150 mm Y = 90 mm Z = 3 mm

Purpose: Electronics lid. Assembly origin: [210, 65, 485] mm.

Native feature sequence: Top Plane: rectangle 150 x 90; Boss-Extrude 3. Fasteners/gasket are pending.

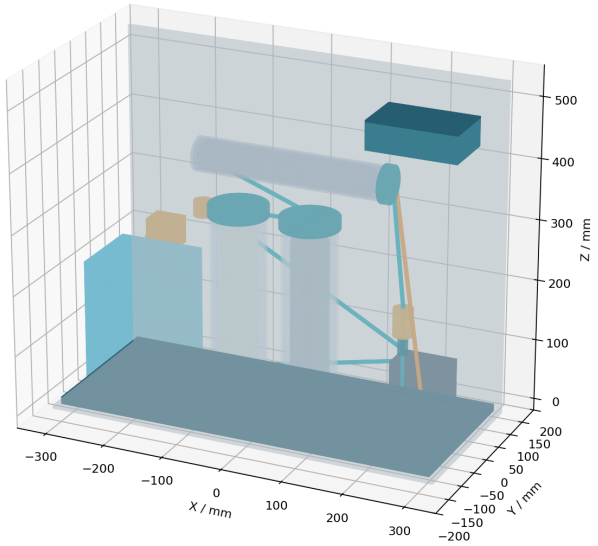
Dimensions define concept envelopes only. No material specification, thread/seal detail, tolerance, certified pressure rating or manufacturing release is implied.

Assembly and routing reference

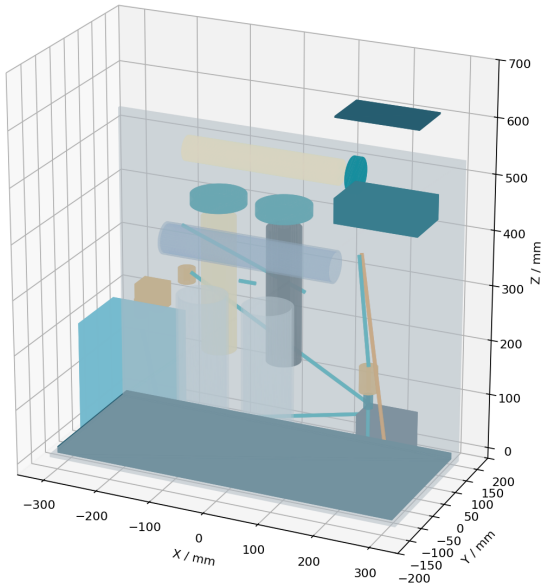
CONCEPT REFERENCE DRAWING | Units: mm | Assumed dimensions | Not for pressure manufacture

Stand: 640 x 280 mm base, 540 mm backplate height. Turquoise tubes: intended supply/product routing; orange: reject routing. Seven tube segments are routing placeholders. Duplicate sensor holders, rated supports, restrictor and collection plumbing are pending detailed integration.

FilterSense / assembled concept
Assumed geometry • Not pressure-rated • Tubing is routing intent



FilterSense / exploded concept
Assumed geometry • Not pressure-rated • Tubing is routing intent



Source geometry: cad/build_cad.py. Assembly placements: parts_manifest.json. Inspect every mating/interface detail before any physical build. Electronic and hydraulic safety are design intent, not validated hardware.